

Assignment 2

Expectations and Random Vectors

Question 1: Randomized Depth-First Search

The joint PMF is a Poisson-Binomial hierarchy: $X_1 \sim \text{Poisson}(\lambda)$ and $X_2 | X_1 \sim \text{Binomial}(X_1, p)$.

(a) Marginals.

For X_1 , sum out x_2 using the binomial theorem $\sum_{x_2=0}^{x_1} \binom{x_1}{x_2} p^{x_2} (1-p)^{x_1-x_2} = 1$:

$$f_{X_1}(x_1) = \frac{e^{-\lambda} \lambda^{x_1}}{x_1!}, \quad x_1 = 0, 1, 2, \dots$$

For X_2 , substitute $x_1 = x_2 + k$ and sum over $k \geq 0$:

$$f_{X_2}(x_2) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^{x_2+k}}{(x_2+k)!} \binom{x_2+k}{x_2} p^{x_2} (1-p)^k = \frac{e^{-\lambda p} (\lambda p)^{x_2}}{x_2!}.$$

So $X_2 \sim \text{Poisson}(\lambda p)$.

(b) Independence of X_2 and $Y = X_1 - X_2$.

Write $x_1 = x_2 + y$. Substituting into the joint PMF and using $\binom{x_2+y}{x_2} = \frac{(x_2+y)!}{x_2! y!}$:

$$f_{X_2, Y}(x_2, y) = \frac{e^{-\lambda} \lambda^{x_2+y}}{(x_2+y)!} \cdot \frac{(x_2+y)!}{x_2! y!} p^{x_2} (1-p)^y = \frac{e^{-\lambda p} (\lambda p)^{x_2}}{x_2!} \cdot \frac{e^{-\lambda(1-p)} (\lambda(1-p))^y}{y!}.$$

This factors into a $\text{Poisson}(\lambda p)$ PMF times a $\text{Poisson}(\lambda(1-p))$ PMF, with separable supports $x_2, y \geq 0$. Therefore X_2 and Y are independent Poisson variables. This is the Poisson splitting theorem.

Question 2: Catching the Greedy Counterfeiter

From each of the n boxes (each holding n coins, m false), one coin is drawn. Each draw is false with probability m/n , independently across boxes. The number of false coins R in the sample is

$$P(R = r) = \binom{n}{r} \left(\frac{m}{n}\right)^r \left(1 - \frac{m}{n}\right)^{n-r}, \quad r = 0, 1, \dots, n.$$

So $R \sim \text{Binomial}(n, m/n)$, with mean m fixed. As $n \rightarrow \infty$ the law of rare events gives

$$P(R = r) \rightarrow \frac{e^{-m} m^r}{r!},$$

so the distribution converges to Poisson(m).

Question 3: Network Server Load

(a) Normalizing constant. Integrate over $0 \leq x_2 \leq x_1 < \infty$:

$$\int_0^\infty \int_0^{x_1} c e^{-x_1} dx_2 dx_1 = c \int_0^\infty x_1 e^{-x_1} dx_1 = c \cdot 1 = 1 \implies c = 1.$$

(b) Conditional PDF. First the marginal of X_1 :

$$f_{X_1}(x_1) = \int_0^{x_1} e^{-x_1} dx_2 = x_1 e^{-x_1}, \quad x_1 > 0.$$

Then

$$f_{X_2|X_1}(x_2 | x_1) = \frac{f(x_1, x_2)}{f_{X_1}(x_1)} = \frac{e^{-x_1}}{x_1 e^{-x_1}} = \frac{1}{x_1}, \quad 0 \leq x_2 \leq x_1.$$

So $X_2 | X_1 = x_1$ is uniform on $[0, x_1]$.

(c) Probability. With $x_1 = 2$, X_2 is uniform on $[0, 2]$:

$$P(X_2 > 1 | X_1 = 2) = \int_1^2 \frac{1}{2} dx_2 = \frac{1}{2}.$$

Question 4: Total Variance in Trading

$S = \sum_{i=1}^N X_i$ is a compound Poisson sum with $N \sim \text{Poisson}(\lambda)$, $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$.

Mean (law of total expectation).

$$E[S] = E[E[S | N]] = E[N\mu] = \mu E[N] = \lambda\mu.$$

Variance (law of total variance).

$$\text{Var}(S) = E[\text{Var}(S | N)] + \text{Var}(E[S | N]) = E[N\sigma^2] + \text{Var}(N\mu) = \lambda\sigma^2 + \mu^2\lambda.$$

Since $\text{Var}(N) = \lambda$ for the Poisson,

$$\text{Var}(S) = \lambda(\sigma^2 + \mu^2) = \lambda E[X^2].$$

Question 5: Uncorrelatedness and Independence

(a) $U = X_1 + X_2$, $V = X_1 - X_2$ are uncorrelated when variances are equal.

$$\text{Cov}(U, V) = \text{Cov}(X_1 + X_2, X_1 - X_2) = \text{Var}(X_1) - \text{Cov}(X_1, X_2) + \text{Cov}(X_2, X_1) - \text{Var}(X_2).$$

The two covariance terms cancel, leaving $\text{Var}(X_1) - \text{Var}(X_2) = 0$ by hypothesis. So U and V are uncorrelated.

(b) Uncorrelated does not imply independent.

Counterexample: let X be symmetric about 0 with $E[X] = 0$, $E[X^3] = 0$ (e.g. $X \sim \text{Uniform}(-1, 1)$), and set $Y = X^2$. Then

$$\text{Cov}(X, Y) = E[X \cdot X^2] - E[X]E[X^2] = E[X^3] - 0 = 0,$$

so they are uncorrelated. But Y is a deterministic function of X , so they are clearly dependent (for instance $P(Y < \frac{1}{4} \mid X = \frac{1}{2}) = 0 \neq P(Y < \frac{1}{4})$). Uncorrelatedness only rules out linear association.

Question 6: Molina's Urns

Let both urns hold the same total T balls. Let urn 1 have white fraction w_1 and urn 2 have white fraction w_2 . Drawing n balls with replacement, the required condition is

$$w_1^n = w_2^n + (1 - w_2)^n.$$

The left side is the probability all n draws from urn 1 are white; the right side is the probability the draws from urn 2 are all white or all black.

Feasibility (real-valued). The right side $g(w_2) = w_2^n + (1 - w_2)^n$ is convex and symmetric about $w_2 = \frac{1}{2}$, with minimum 2^{1-n} at $w_2 = \frac{1}{2}$ and supremum approaching 1 at the endpoints. So $g(w_2) \in [2^{1-n}, 1)$. The left side w_1^n ranges over $(0, 1)$. The ranges overlap, so real solutions exist: for example with $n = 3$ and $w_2 = \frac{1}{2}$, the right side is $\frac{1}{4}$, giving $w_1 = (1/4)^{1/3} \approx 0.63$.

Integer compositions. With genuinely mixed urns (both colors present, $0 < w_1, w_2 < 1$) a brute-force search over all totals T up to several hundred and all $n \geq 3$ finds no exact rational solution: the equality $w_1^n = w_2^n + (1 - w_2)^n$ forces w_1 to be irrational in essentially every case. The only exact solution with integer compositions is degenerate: urn 1 all white ($w_1 = 1$) and urn 2 all black ($w_2 = 0$), where both sides equal 1.

Conclusion. It is possible in the real-valued sense (and the curve of valid (w_1, w_2) pairs is described above), but not achievable with non-degenerate integer urn compositions. The clean nontrivial answer requires irrational white fractions, which no finite urn can realize exactly.